

Markscheme

Mathematics: analysis and approaches
Practice question bank

90. (a)

The expression $n^2 + n$ can be factorised as $n(n + 1)$, the product of two consecutive integers. Since one of any two consecutive integers must be even, this proves $n^2 + n$ is always even, for every integer n .

Using this result, state how many of the values $n^2 + n$, for $n = 1, 2, 3, \dots, 100$, are odd.

since the proof shows $n^2 + n$ is even for every integer n , none of these 100 values can be odd (M1) R1

the number of odd values is 0 A1

[4 marks]